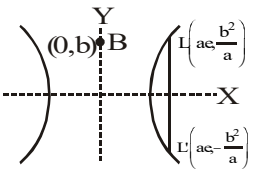
SOLUTION

SECTION–I

1. Ans.(C)

Given that $m_{BL}.m_{BL'} = -1$



$$\left(\frac{\frac{b^2}{a}-b}{ae}\right).\left(\frac{-\frac{b^2}{a}-b}{ae}\right) = -1$$

$$\Rightarrow \quad \frac{b^4}{a^2} - b^2 = a^2e^2$$

$$\Rightarrow \quad \frac{b^4}{a^4} - \frac{b^2}{a^2} = e^2$$

$$\Rightarrow \quad (e^2-1)^2 - (e^2-1) = e^2$$

$$\Rightarrow \quad e^4 - 4e^2 + 2 = 0$$

$$\Rightarrow \quad e^2 = 2 \pm \sqrt{2}$$

$e^2 = 2 + \sqrt{2}$ for hyperbola.

2. Ans. (B)

$\because$ slope of chord joining $A(t_1), B(t_2) = \frac{2}{t_1 + t_2}$

and slope of nomral at $A(t_1) = -t_1$

$$\Rightarrow m = \frac{2}{t_1 + t_2} = -t_1$$

Also $t_1t_2 = -4$

$\therefore \ t_1 = \pm\sqrt{2}$

3. Ans. (B)

$\because$ focus (2,0) lies on the line $\ell x - y - 2\ell = 0$ hence it is focal chord.

Tangents drawn at end points of a focal chord intersect at right angle.

4. Ans. (A)

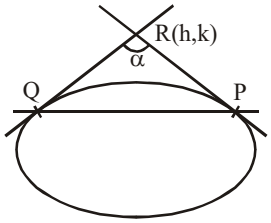
For tangent use : $C^2 = A^2m^2 + B^2$

$c^2 = 4c^2 + 1$

$\Rightarrow \quad 3c^2 + 1 = 0$

Not possible

5. Ans. (D)



Chord of contact of R

$hx + 2ky = 6$

compare with $\frac{x}{2}\cos\theta + y\sin\theta = 1$

$$\frac{2h}{\cos\theta} = \frac{2k}{\sin\theta} = 6$$

$h^2 + k^2 = 9$

which is director circle of $x^2 + 2y^2 = 6$.

6. Ans. (A)

Equation of chord with given middle point (x, y) is

$$T = S_1 \Rightarrow \frac{xx_1}{4} + \frac{yy_1}{9} = \frac{x_1^2}{4} + \frac{y_1^2}{9} \quad ...(1)$$

Compare (1) with $9x - 4y = 13$

$x_1 = 1$ and $y_1 = -1$ (i.e. $\alpha = 1$ & $\beta = -1$)

Hence, $(\alpha + \beta) = 0$

7. Ans. (A,B,C,D)

Given minor axis $\Rightarrow$ $2y + x = 3$

Equation of major axis = $2x - y = k$

Passes throu (2, 3) $\quad k = 1$

hence major axis $2x - y = 1$

centre of ellipse (1, 1)

$$\frac{a}{e} = \sqrt{(3-1)^2 + (2-1)^2} = \sqrt{5}$$

$\because \ b^2 = a^2(1 - e^2)$

$$\therefore \ \frac{5}{4} = 5e^2(1-e^2) \Rightarrow e = \frac{1}{\sqrt{2}}$$

$$a = \frac{\sqrt{5}}{\sqrt{2}}$$

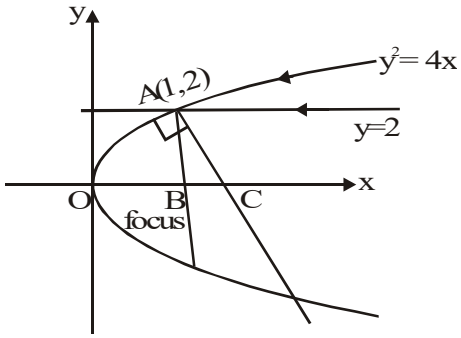
equation of ellipse

$$4(2x - y - 1)^2 + 2(x + 2y - 3)^2 = 25$$

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8. Ans.(B,D)



Let A $(t^2, 2t)$

$\because$ y = 2 $\therefore$ 2t = 2

t = 1

again meet C $(t_1^2, 2t_1)$

equation of normal at A

y + x = 3

put y = 0, x = 3

C ≡ (3, 0)

B is focus (1, 0)

9. Ans. (B,C,D)

Let $P_1(a\sec\theta, b\tan\theta)$ be a point on

$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$$

Now chord of contact w.r.t. P_1 to the ellipse is

$$\frac{x\sec\theta}{a}+\frac{y\tan\theta}{b}=1$$

Now solve with $y=\pm\frac{b}{a}x$

$$\frac{x}{a}\sec\theta+y\frac{\tan\theta}{a}=1$$

Q(a(secθ – tanθ), b(secθ – tanθ))

Similarly $\frac{x\sec\theta}{a}+\frac{y\tan\theta}{b}=1$

and $y=-\frac{b}{a}x$

x = a(secθ + tanθ)

R(a(secθ + tanθ), – b(secθ + tanθ))

Mid point of Q and R is

h = a secθ and k = – b tanθ

$$\Rightarrow \frac{h^2}{a^2}-\frac{k^2}{b^2}=1$$

(C) For mirror image w.r.t. x axis

Replace y → –y

Equation remains same.

(D) Locus of foot of perpendicular to the line QR from (ae,0) is auxilliary circle. As line QR itself is tangent to the given hyperbola.

10. Ans. (B,C,D)

$$\Rightarrow \quad (\log b)^2=(\log a)^2\left[1-\frac{1}{2}\right]$$

$$2=\left(\log_b a\right)^2$$

$$e^2=1+\left(\log_a b\right)^2=1+\frac{1}{2}=\frac{3}{2}$$

11. Ans. (B,D)

C is $\frac{x^2}{7}-\frac{y^2}{3}=21 \ (y\geq 0)$

$\because$ Equation of L will be y = x – 2 and $A\left(\frac{7}{2},\frac{3}{2}\right)$

Equation of circle

$$\left(x-\frac{7}{2}\right)^2+\left(y-\frac{3}{2}\right)^2+\lambda(x-y-2)=0$$

passing through (0, 0) $\Rightarrow$ $\lambda=\frac{29}{4}$

12. Ans. (C,D)

Let $P\left(\frac{\sqrt{325}}{2},y_1\right)$

$$\frac{325}{25\times 4}-\frac{y_1^2}{16}=1$$

$$\frac{y_1^2}{16}=\frac{225}{100}$$

$$y_1=\pm\frac{15\times 4}{10}=\pm 6$$

Let $P\left(\frac{\sqrt{325}}{2},-6\right)$

Tangents is $\frac{x.5.\sqrt{13}}{2\times 25}+\frac{6y}{16}=1$

$$\frac{x.\sqrt{13}}{10}+\frac{3y}{8}=1$$

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$$x = \left(1 - \frac{3y}{8}\right) \times \frac{10}{\sqrt{13}}$$

Solve with $x^2 + y^2 = 25$

$$\frac{100}{13}\left(1 - \frac{3y}{8}\right)^2 + y^2 = 25$$

$$100\left(1 - \frac{3y}{8}\right)^2 + 13y^2 = 325$$

$$y^2\left(13 + \frac{900}{64}\right) - \frac{600y}{8} - 225 = 0$$

$$\text{H.M.} = \left|\frac{2\delta\gamma}{(\gamma+\delta)}\right| = \left|\frac{2(-225)}{\left(\frac{600}{8}\right)}\right|$$

$$= \frac{16 \times 15}{40} = |-6| = 6$$

Paragraph for Question 13 and 14

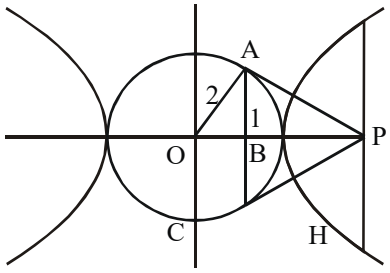
$$\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1 \qquad \therefore\ e_1 = e_2$$

$\Rightarrow \quad e_1 = \sqrt{2} \quad \Rightarrow a = b$

foot of perpendicular drawn from focus upon any tangent always lies on auxiliary circle.

$x^2 + y^2 = a^2 \quad \Rightarrow \quad a = 2 \ \Rightarrow \ b = 2$

$C : x^2 + y^2 = 4$ (which is auxiliary circle itself)



13. Ans. (A)

Number of common tangent of H & C is 2

14. Ans. (C)

If ellipse & hyperbola cuts orthogonally
$\Rightarrow$ confocal.

foci of H $\Rightarrow \quad (\pm 2\sqrt{2}, 0)$

$\Rightarrow (4e_e, 0) = (2\sqrt{2}, 0)$

$$e_e = \frac{1}{\sqrt{2}}$$

$$\beta^2 = 16\left(1 - \frac{1}{2}\right) = 8 \Rightarrow \beta = 2\sqrt{2}$$

Paragraph for Question 15 and 16

15. Ans. (D)

Let A $(t_1^2, 2t_1)$ and B$(t_2^2, 2t_2)$

$\because$ AB is focal chord $\therefore\ t_1 t_2 = -1$

Now $t_1 = \frac{y_1}{2}, t_2 = \frac{y_2}{2}$

$$\Rightarrow \frac{y_1}{2}.\frac{y_2}{2} = -1 \ \ \& \ \ x_0 = \frac{x_1 + x_2}{2}, y_0 = \frac{y_1 + y_2}{2}$$

$$A\left(\frac{y_1^2}{4}, y_1\right)$$

$$B\left(\frac{y_2^2}{4}, y_2\right) \equiv \left(\frac{4}{y_1^2}, \frac{-4}{y_1}\right)$$

$$C\left(\frac{y_3^2}{4}, y_3\right)$$

$$D\left(\frac{y_4^2}{4}, y_4\right)$$

$$M_{BC} = \frac{y_3 - y_2}{x_3 - x_2}$$

$$m_{AE} = \frac{y_1 - 0}{x_1 - 2}$$

$\because \quad \frac{y_3 - y_2}{x_3 - x_2} = \frac{y_1}{x_1 - 2}$

$$\frac{4}{y_3 + y_2} = \frac{4y_1}{y_1^2 - 8}$$

$\Rightarrow \quad 4\left(y_1^2 - 8\right) = 4y_1y_3 + 4(-4)$

$\Rightarrow \quad y_3 = \frac{4y_1^2 - 16}{4y_1} = y_1 - \frac{4}{y_1} = y_1 + y_2 = 2y_0$

16. Ans. (B)

Parabola $y^2 = 4x$ & $y_1y_4 = 4$

tangent at D : $yy_4 = 2(x + x_4)$

normal at A : $y - y_1 = \frac{-y_1}{2}(x - x_1)$

$\Rightarrow \quad yy_4 - y_1y_4 = \frac{-y_1y_4}{2}(x - x_1)$

abssissa of intersection point is

$$x = 1 + \frac{x_1 - x_4}{2}$$

$$= 1 + \frac{x_1 - \frac{1}{x_1}}{2} \quad \left\{\text{Use}: \frac{1}{x_1} = x_2 \ ; \ 2x_0 = x_1 + x_2\right\}$$

$$= 1 + x_0 - x_2$$

17. Ans. (A)

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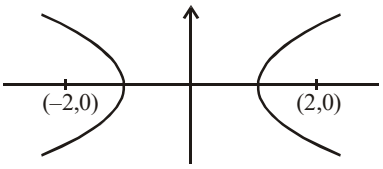
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For hyperbola 'H'

$ae = 2,\ e = 2 \Rightarrow a = 1$

$b^2 = 3$

$\frac{x^2}{1} - \frac{y^2}{3} = 1,\ LR = 6$



18. Ans. (D)

Let $y = mx + c$ is common tangent

$\therefore\ c = -m^2 = \pm\sqrt{\frac{m^2}{4} + \frac{3}{4}} = \frac{a}{m}$

$\Rightarrow m = -1,\ a = 1,\ c = -1$

common tangent $y = -x - 1$

$x + y + 1 = 0$

$\ell = 1,\ m = 1$

SECTION–II

1. Ans. 4.50

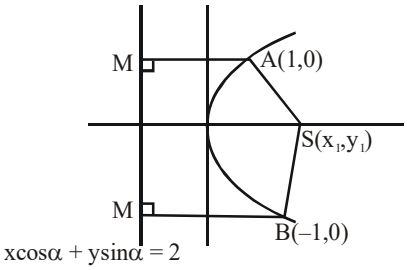
Tangent to parabola $y = mx + \frac{2}{m}$

is also tangent to hyperbola.

$\therefore\ c^2 = a^2m^2 - b^2 \Rightarrow \frac{4}{m^2} = m^2 - 3 \Rightarrow m = \pm 2$

$\therefore$ Area $= \frac{1}{2}.6.\frac{3}{2} = \frac{9}{2}$

2. Ans. 1.33 or 1.34



Let directrix touches the circle at $(2\cos\alpha, 2\sin\alpha)$

$\Rightarrow$ equation of directix is $x\cos\alpha + y\sin\alpha = 2$

Let focus be $S(x_1, y_1)$

A(1,0), B(–1,0) lies on parabola so

$(x_1 - 1)^2 + y_1^2 = (\cos\alpha - 2)^2$ ...(1)

(for parabola, PS = PM, P is considered as A(1,0))

$(x_1 + 1)^2 + y_1^2 = (\cos\alpha + 2)^2$ ...(2)

(for parabola, PS = PM, P is considered as B(–1,0))

$x_1 = 2\cos\alpha\ \ y_1 = \sqrt{3}\sin\alpha$

Locus of focus $\Rightarrow \frac{x^2}{4} + \frac{y^2}{3} = 1$

3. Ans. 4.00

$AS_1 + AS_2 \geq 2\sqrt{AS_1.AS_2}$

$4 \geq 2\sqrt{AS_1.AS_2}$ $(\because\ AS_1 + AS_2 = 2a = 4)$

$AS_1.AS_2 \leq 4$

4. Ans. 2.00

Let point on line $\left(h, \frac{1}{t+2} - h\right)$

chord of contact $4xh + 16y\left(\frac{1}{t+2} - h\right) = 1$

it passes through fixed point so $x = 4y$ &

$\frac{16y}{t+2} - 1 = 0$

fixed point $= \left(\frac{t+2}{4}, \frac{t+2}{16}\right)$ lies inside ellipse

So $S_1 < 0$

$-\frac{4}{\sqrt{5}} < (t+2) < \frac{4}{\sqrt{5}}$

$t = -1, -3,\quad (t \neq -2)$

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